

BUSINESS PROBLEM

Customer Shopping Behavior Analysis · Project Brief

01 | Background & Business Need

A major retail business aims to gain a clearer picture of how customers shop so it can increase sales, strengthen customer satisfaction, and encourage repeat business. The team has observed that buying habits vary across customer groups, product types, and shopping channels, including online and in-store purchases.

The business also wants to understand how factors such as promotional discounts, product ratings and reviews, seasonal changes, and preferred payment methods influence purchase decisions and customer loyalty.

02 | Core Business Question

How can customer shopping data be used to uncover meaningful trends, improve customer engagement, and make marketing and product planning more effective?

03 | Project Deliverables

01	Data Preparation & Modeling — Python Clean, organize, and transform the raw customer dataset so it is ready for reliable analysis.
02	Data Analysis — SQL Structure the data, represent business transactions where needed, and query it to discover customer segments, loyalty patterns, and factors linked to purchase decisions.
03	Dashboard & Insights — Power BI Develop an interactive dashboard that presents key patterns and trends and helps stakeholders make informed, data-driven decisions.
04	Project Report & Presentation Document the main findings and business recommendations, then communicate the insights clearly through a visual presentation.
05	GitHub Repository Publish the project files in a well-organized repository, including Python scripts, SQL queries, and dashboard files.

PROJECT FOCUS

Turn raw shopping data into practical insights that support better customer and business decisions.